HERIOT WATT UNIVERSITY

# Multivariable Calculus and Analysis A Worksheet 1

## Starter Questions

**1)** (a) $|x-2| = 10 \Rightarrow x - 2 = \pm 10$. Hence, $x = -8$ or $x = 12$.

(b) $|x-4| < 7 \Rightarrow -7 < x - 4 < 7 \Rightarrow -3 < x < 11$, where the first implication follows from Theorem 3.3 (vii).

(c) $|x+2| < 1 \Rightarrow -1 < x + 2 < 1 \Rightarrow -3 < x < 1$, where the first implication follows from Theorem 3.3 (vii).

(d) $|x-1||x+1| = 0 \Rightarrow x - 1 = 0$ or $x + 1 = 0$. Hence, $x = -1$ or $x = 1$. The first implication follows from Theorem 3.3 (i).

**2)** *Proof.* Let $a, b \in \mathbb{R}$. Then by the triangle inequality it follows that

$$|a| = |a - b + b| \leq |a - b| + |b|.$$

Hence, $|a| - |b| \leq |a - b|$. Similarly, by the triangle inequality if follows that

$$|b| = |b - a + a| \leq |b - a| + |a|.$$

By Theorem 3.3 (ii), $|b - a| = |a - b|$ and hence

$$|b| - |a| \leq |a - b| \Rightarrow |a| - |b| \geq -|a - b|.$$

Putting these inequalities together gives

$$-|a - b| \leq |a| - |b| \leq |a - b|.$$

By Theorem 3.3 (vii) it follows that

$$||a| - |b|| \leq |a - b|.$$

□

**3)** (a) Since $m < n$ it follows that $m/n < 1$ for all $m, n \in \mathbb{N}$ and thus the set is bounded above. Similarly, since $m, n > 0$ the set is also bounded below. So by the completeness axiom both the supremum and infimum must exist. After some thought, the infimum of the set is given by $0$ and the supremum is $1$. Neither of these elements are contained in the set and hence there is no maximum or minumum of the set.

(b) By observing that

$$\frac{(-1)^m}{n} = \begin{cases} 1/n, \ m \text{ is even} \\ -1/n, \ m \text{ is odd} \end{cases}$$

it follows that the set is bounded above by $1$ and below by $-1$. Hence, by the completeness axiom both the infimum and supremum exists. The supremum is given by $1$ and infimum by $-1$. In this case, since both $-1$ and $1$ are also in the set, they are the minumum and maximum of the set respectively.

(c) By observing that

$$\frac{n}{3n+1} = \frac{1}{3}\left(\frac{3n+1-1}{3n+1}\right) = \frac{1}{3}\left(1 - \frac{1}{3n+1}\right)$$

it follows that the set is bounded above by $1$ and below $0$. Hence, by the completeness axiom both the supremum and infimum exist. In this case, the supremum is given by $1/3$ and the infimum by $1/4$. Since $1/4$ is in the set, this is the minumum of the set. However, $1/3$ is not in the set and hence the set has no maximum.

(d) Since $m, n \in \mathbb{N}$ it is clear that $m/(m+n) > 0$ and thus the infimum exists. Letting $m = 1$, gives $1/(1+n)$ which clearly has limit $0$ as $n$ tends to infinity. Hence, the infimum of the set is $0$. Now, $m < m + n$ for all $m, n \in \mathbb{N}$ and thus $m/(m+n) < 1$. By the completeness axiom, the supremum of the set exists. The supremum would then be the same as the one given in part (a), namely $1$. Neither $1$ nor $0$ are in the set and hence the maximum and minumum do not exist.

**4)** (a) This statement is false. One counterexample is the set $\mathbb{N} \subset \mathbb{R}$. Since $\mathbb{N}$ is not bounded above, which follows from the Archimedean principle, it doesn't have a supremum.

(b) The statement is true.

*Proof.* If $M \in \mathbb{R}$ is an upper bound for the set $A$ then $a \leq M$ for all $a \in A$. Now if $L \in \mathbb{R}$ satisfies $M < L$ it follows that $a < L$ for all $a \in A$. Hence, $L$ is an upper bound for $A$. □

**5)** This statement is false. One counterexample is if $A = \{-1, 0, 1\}$ and $B = \{-1, 1\}$. In this case, $\sup(A) = \sup(B)$ and $\inf(A) = \inf(B)$ but clearly $A \neq B$ since $0 \in A$ and $0 \notin B$.

**6)** (a) The condition that defines the elements of set is equivalent to

$$3x^2 - 10x + 3 < 0.$$

Factorising this equation gives

$$(3x - 1)(x - 3) < 0.$$

Hence, the set is equal to the open interval $(1/3, 3)$. It follows then that the supremum of the set is $3$ and the infimum is $1/3$. Since neither of these elements are contained in the set, the maximum and minimum do not exist.

(b) From the previous part, it follows that this set is simply all the integers contained in the interval $(1/3, 3)$. That is, the set is equal to $\{1, 2\}$. Hence, the supremum of the set is $2$ and the infimum is $1$. Unlike before, both of these elements are contained in the set and thus $1$ is the minimum and $2$ is the maximum.

**7)** (a) *Proof.* Assume towards a contradiction that $\sqrt{3}$ is rational. That is, $\sqrt{3} = p/q$ with $p/q$ irreducible. Squaring both sides gives

$$3q^2 = p^2.$$

By *the fundamental theorem of arithmetic*, it follows that if $3 \mid p^2$ then $3 \mid p$. Hence it follows that $9 \mid p^2$ which means $3 \mid q^2$. By the same argument as before, $3 \mid q$. So both $3 \mid p$ and $3 \mid q$ hence $p/q$ is reducible. This is a contradiction and thus $\sqrt{3}$ is irrational. □

(b) Where the proof that $\sqrt{4}$ is irrational would fail is in the statement $4 \mid p^2 \Rightarrow 4 \mid p$. This is not true if the divisor is a perfect square. Since $4 = 2^2$, and hence a perfect square, the statement is not true. One explicit example would be that $4 \mid 36$ but $4 \nmid 6$.

**8)** *Proof.* ($\Rightarrow$) Assume $\beta = \inf(B)$. Then by definition $\beta$ is the greatest lower bound for $B$ and hence $\beta + \varepsilon$ is not a lower bound for $B$ for every $\varepsilon > 0$. Thus, there must exist a $b \in B$ such that $b < \beta + \varepsilon$.

($\Leftarrow$) Assume that $\beta \in \mathbb{R}$ is a lower bound for $B$ and for every $\varepsilon > 0$, there exists a $b \in B$ such that $b < \beta + \varepsilon$. This means that any number greater than $\beta$ is not a lower bound for $B$. Thus any other lower bound $c \in \mathbb{R}$ for $B$ must satisfy $c \leq \beta$. Hence $\inf(B) = \beta$. □

**9)** *Proof.* Suppose towards a contradiction that there exists two distinct infima for $A$, say $\beta, \beta' \in \mathbb{R}$. Since the infimum is the greatest lower bound, it follows that both $\beta \leq \beta'$ and $\beta' \leq \beta$. Hence $\beta = \beta'$. This is a contradiction and it follows that infima are unique. □

## Challenge Questions

**10)** (a) *Proof.* By the triangle inequality it follows that

$$|2a + 5b - 7L| = |2(a - L) + 5(b - L)| \leq 2|a - L| + 5|b - L| < 7r.$$

□

(b) *Proof.* Similarly, by the triangle inequality it follows that

$$|3a - 2b - L| = |3(a - L) - 2(b - L)| \leq 3|a - L| + 2|b - L| < 5r.$$

□

**11)** *Proof.* Expanding $|x^2 - 16| < r$ gives

$$|x^2 - 16| = |x - 4||x + 4| < \delta|x + 4| \leq \frac{r}{9}|x + 4|.$$

Now, $|x - 4| < \delta \leq 1$. This means that $-1 < x < 5 \Rightarrow |x + 4| < 9$. Piecing these inequalities together gives

$$|x^2 - 16| < \frac{r}{9}|x + 4| < \frac{9r}{9} = r.$$

□

**12)** (a) *Proof.* Let $a = p/q \in \mathbb{Q}$ and $b = m/n \in \mathbb{Q}$. It follows from the definition of addition and multiplication on $\mathbb{Q}$ that

$$a + b = \frac{pn + mq}{nq}$$

and

$$ab = \frac{pm}{qn}.$$

Since $\mathbb{Z}$ is closed under addition and multiplication if follows that $a + b \in \mathbb{Q}$ and $ab \in \mathbb{Q}$. □

(b) *Proof.* Assume towards a contradiction that $a \in \mathbb{Q}$, $r \in \mathbb{I}$ and $a + r \in \mathbb{Q}$. It follows then that $a + r = b$ for some $b \in \mathbb{Q}$. Rearranging gives $r = b - a$. By part (a), $\mathbb{Q}$ is closed under addition so $r \in \mathbb{Q}$. This is a contradiction. Hence $r + a \in \mathbb{I}$. □

*Proof.* Assume towards a contradiction that $a \in \mathbb{Q} \setminus \{0\}$, $r \in \mathbb{I}$ and $ar \in \mathbb{Q}$. It follows that $ar = b$ for some $b \in \mathbb{Q}$. Rearranging gives $r = a^{-1}b$ and since $a^{-1} \in \mathbb{Q}$ and $\mathbb{Q}$ is closed under multiplication, by part (a), it follows that $r \in \mathbb{Q}$. This is a contradiction and thus $ar \in \mathbb{I}$. □

(c) Both of these statements are false. A simply counter example to both is if $r = \sqrt{2}$ and $t = -\sqrt{2}$. It follows that $r + t = 0 \notin \mathbb{I}$ and $rt = -2 \notin \mathbb{I}$.

**13)** *Proof.* Let $x < y$ with $x, y \in \mathbb{R}$. It follows that $x - \sqrt{2} < y - \sqrt{2}$ and $x - \sqrt{2}, y - \sqrt{2} \in \mathbb{R}$. By the density of the rationals in the reals, there exist $z \in \mathbb{Q}$ such that

$$x - \sqrt{2} < z < y - \sqrt{2}$$
$$\Rightarrow x < z + \sqrt{2} < y.$$

By question 12 part (b), it follows that $z + \sqrt{2} \in \mathbb{R} \setminus \mathbb{Q}$. Thus the irrational numbers are dense in the reals. □

**14)** *Proof.* Let $A \subset \mathbb{R}$ be non-empty and bounded below. That is, there exists a $x \in \mathbb{R}$ such that $x \leq a$ for all $a \in A$. Define

$$B := -A = \{-a : a \in A\}.$$

Since $x \leq a$ it follows that $-a \leq -x$ for all $a \in A$. Hence, the set $B$ is bounded above and by the completeness axiom has a supremum. Call $\sup(B) = -\beta \in \mathbb{R}$. Then $-a \leq -\beta \leq -x$ which implies $x \leq \beta \leq a$ for all $a \in A$. Hence $\beta$ is the greatest lower bound for $A$ and thus $\inf(A) = -\sup(B) = \beta$. □

**15)** If open intervals were used instead of closed internals in the nested interval theorem it wouldn't be the case that there exists a $x \in I_n$ for every $n \in \mathbb{N}$. For example, take

$$I_n = \left(0, \frac{1}{n}\right).$$

There is no number $x \in \mathbb{R}$ that is an element of $I_n$ for each $n \in \mathbb{N}$. If there were, this would contradict the Archimedean property.

**16)** *Proof.* Since $a < 2$ for each $a \in A$ it follows that $2$ is an upper bound for $A$. By the density of the rationals, for every $\varepsilon > 0$ there exists a $r \in \mathbb{Q}$ such that $2 - \varepsilon < r < 2$. Since $r \in A$ it follows by Theorem 4.2 that $\sup(A) = 2$. □

**17)** (a) *Proof.* As $n \in \mathbb{N}$ it follows that $1/n > 0$ and $1 + 1/n > 1$. Hence $1$ is a lower bound for the set. For any $\varepsilon > 0$ it follows by the Archimedean property there exist an $N \in \mathbb{N}$ such that $N > 1/\varepsilon$. Hence $1 + 1/N < 1 + \varepsilon$ and $1 + 1/N$ is contained in the set. It follows by Theorem 4.2 (ii) that the infimum of the set is $1$. □

(b) *Proof.* Observing that

$$\frac{n}{n+1} = \frac{n+1-1}{n+1} = 1 - \frac{1}{n+1}$$

it follows that

$$\left\{\frac{n}{n+1} : n \in \mathbb{N}\right\} = \left\{1 - \frac{1}{n} : n \in \mathbb{N}\right\} \setminus \{0\}.$$

It was proven in lecture that $\sup(\{1 - 1/n : n \in \mathbb{N}\}) = 1$. Since $1 > 0$, it follows that

$$\sup\left(\left\{\frac{n}{n+1} : n \in \mathbb{N}\right\}\right) = 1.$$

□

(c) *Proof.* Similar to part (b), the set can be rewritten as

$$\left\{\frac{n}{n+1} : n \in \mathbb{N}\right\} = \left\{1 - \frac{1}{n} : n \in \mathbb{N} \setminus \{1\}\right\}.$$

Since $n \geq 2$ it follows that $-1/n \geq -1/2$. Hence $1/2 \leq 1 - \frac{1}{n}$ for all $n \in \mathbb{N} \setminus \{1\}$. That is, $1/2$ is a lower bound for the set. By question 14, the infimum of the set exists. Assume towards a contradiction that the infimum of the set is $\beta \neq 1/2$. Then there exists an $\varepsilon > 0$ such that for every $N \in \mathbb{N} \setminus \{1\}$ it follows that

$$\frac{1}{2} + \varepsilon \leq 1 - \frac{1}{N}.$$

This is equivalent to

$$\varepsilon \leq \frac{1}{2} - \frac{1}{N}.$$

For $N = 2$, the gives

$$\varepsilon \leq 0$$

which is a contradiction since $\varepsilon > 0$. Hence $\beta = 1/2$. □

## Extension Questions

**18**) *Proof.* Suppose towards a contradiction that $\sqrt{k}$ is not an integer and is rational. Then there exist $a, b \in \mathbb{N}$ such that

$$\sqrt{k} = \frac{a}{b},$$

with $a/b$ irreducible. Let $n$ be the largest integer less than $\sqrt{k}$, so that $0 < \sqrt{k} - n < 1$.

Following the steps given,

$$\begin{aligned}\sqrt{k} &= \frac{a}{b} \\ &= \frac{a\left(\sqrt{k}-n\right)}{b\left(\sqrt{k}-n\right)} \\ &= \frac{a\sqrt{k}-an}{b\sqrt{k}-bn}.\end{aligned}$$

Since $\sqrt{k} - n < 1$, the numerator and denominator of the right-hand side are less than $a$ and $b$ respectively. Substitute $a = b\sqrt{k}$ into the numerator, and $\sqrt{k} = a/b$ into the denominator to obtain

$$\begin{aligned}\sqrt{k} &= \frac{b\sqrt{k}\sqrt{k}-an}{b(a/b)-bn} \\ &= \frac{bk-an}{a-bn}.\end{aligned}$$

Recall that $a, b, k, n \in \mathbb{N}$. Also recall that the numerator of the expression on the right-hand side is positive and less than $a$, and the denominator is positive and less than $b$. Hence, there exist integers $a'$ and $b'$ such that

$$\begin{aligned}a' &= bk - an &&< a, \\ b' &= a - bn &&< b, \text{ and} \\ \sqrt{k} &= \frac{a'}{b'}.\end{aligned}$$

The process above can be repeated indefinitely to produce positive integers $\widetilde{a}$ and $\widetilde{b}$ that are successively smaller after each iteration. However, there exists a smallest positive integer, $1$. This is a contradiction and thus $\sqrt{k}$ is irrational.

It follows then that if $\sqrt{k}$ is not an integer, then it is irrational. □

**19**) *Proof.* Let $n \in \mathbb{N}$ and $x > 0$. Define

$$S = \{y \in \mathbb{R} : y > 0 \text{ and } y^n < x\}.$$

We wish to show $S$ is non-empty and bounded above. To show that $S$ is non-empty it suffices to show there exists an element $s \in \mathbb{R}$ such that $s > 0$ and $s^n < x$. Consider the number

$$s = \frac{x}{x+1}.$$

Since $x > 0$, it follows that $s > 0$, $s < 1$ and $s < x$. Thus $s^n < s < x$. which means $S$ is non-empty. Now consider the number $u = 1 + x$. Since $u > 1$ it follows that $u^n > u$ and thus $u^n > x$. This means that $u \notin S$ but for every $y \in S$, it must be that $y < u$. Hence $S$ is is bounded above and by the completeness axiom its supremum exists. Denote $\sup(S) = \alpha$. Before moving further, it is useful to observe the inequality

$$b^n - a^n < (b - a)nb^{n-1}$$

for all $0 < a < b$ and $n \in \mathbb{N}$. This follows from the factorisation

$$b^n - a^n = (b - a)(b^{n-1} + b^{n-2}a + \cdots + a^{n-1}).$$

Suppose towards a contradiction that $\alpha^n < x$. Let

$$\varepsilon = \min\left\{\frac{1}{2}, \frac{x - \alpha^n}{n(\alpha + 1)^{n-1}}\right\}.$$

Note that since it was assumed that $\alpha^n < x$ this means $0 < x - \alpha^n$ and thus $\varepsilon > 0$. This choice of $\varepsilon$ will be apparent in the next calculation. It follows now that

$$\begin{aligned}(\alpha + \varepsilon)^n - \alpha^n &< \varepsilon n(\alpha + \varepsilon)^{n-1} \\ &< \varepsilon n(\alpha + 1)^{n-1} \\ &\leq x - \alpha^n.\end{aligned}$$

Hence, $\alpha^n < (\alpha + \varepsilon)^n < x$. This is a contradiction as $\alpha$ was an upper bound for $n$.

Now assume towards a contradiction that $\alpha^n > x$. Define

$$\varepsilon = \frac{\alpha^n - x}{n\alpha^{n-1}}.$$

This time, since it is assumed that $\alpha^n > x$ it follows $\alpha^n - x > 0$ and thus $\varepsilon > 0$. A short calculation then shows that

$$\begin{aligned}\alpha^n - (\alpha - \varepsilon)^n &< \varepsilon n\alpha^{n-1} \\ &= \alpha^n - x.\end{aligned}$$

Thus, $x < (\alpha - \varepsilon)^n$ and thus $\alpha - \varepsilon < \alpha$ is an upper bound for $S$. This is a contradiction since $\alpha$ is the least upper bound. It must be the case that $\alpha^n = x$. All that remains is to prove that there is only one such $\alpha > 0$ such that $\alpha^n = x$. Assume towards a contradiction that there exists an $\widetilde{\alpha} \neq \alpha$ such that $\widetilde{\alpha}^n = \alpha^n = x$. Then

$$\begin{aligned}0 &= \alpha^n - \widetilde{\alpha}^n \\ &= (\alpha - \widetilde{\alpha})(\alpha^{n-1} + \alpha^{n-1}\widetilde{\alpha} + \cdots + \widetilde{\alpha}^{n-1}).\end{aligned}$$

Since $\alpha, \widetilde{\alpha} > 0$ it follows that $\alpha^{n-1} + \alpha^{n-1}\widetilde{\alpha} + \cdots + \widetilde{\alpha}^{n-1} > 0$. Hence, by the above expression, $\alpha = \widetilde{\alpha}$. This is a contradiction and it follows that there must only exist one positive number $\alpha \in \mathbb{R}$ such that $\alpha^n = x$. $\square$